

Q1.

- (a) Ultrasound is sound above the maximum frequency that humans can hear.

Tick (✓) **one** box.

20 Hz

☐

2000 Hz

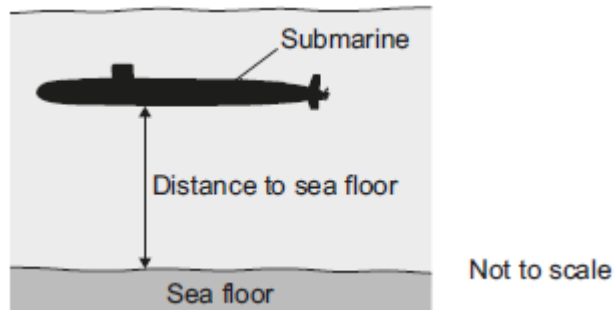
☐

20 000 Hz

☐

(1)

- (b) The image shows a submerged submarine.



The submarine sends a pulse of ultrasound to the sea floor.
The pulse takes 0.25 seconds to travel from the submarine to the sea floor.

The speed of sound in water is 1600 m/s.

Calculate the distance from the submarine to the sea floor.

Distance = _____ m

(2)

- (c) The ultrasound is reflected from the sea floor back to the submarine.
Use the correct answer from the box to complete the sentence.

half	the same as	twice
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The total distance the ultrasound pulse travelled is _____ the distance to the sea floor.

(1)

- (d) The submarine moves through the sea and every few seconds sends a pulse of ultrasound to check the distance to the sea floor.

The table shows the time taken for five ultrasound pulses to travel from the submarine to the sea floor and back to the submarine.

Pulse number	Time for pulse to return in seconds
1	0.50
2	0.45
3	0.38
4	0.40
5	0.48

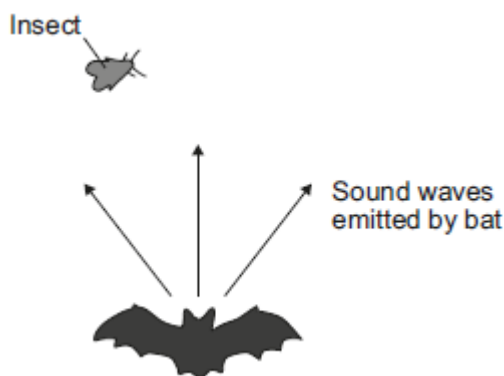
Describe how the distance from the submarine to the sea floor changed over these five pulses.

(2)

(Total 6 marks)

Q2.

Bats use the reflection of high pitched sound waves to determine the position of objects. The image below shows a bat and an insect flying in front of the bat.



- (a) What determines the pitch of a sound wave?

Tick (✓) **one** box.

	Tick (✓)
amplitude	

frequency	
speed	

(1)

- (b) State the name given to reflected sound waves.

(1)

- (c) The bat emits a sound wave with a frequency of 25.0 kHz and a wavelength of 0.0136 metres.

Calculate the speed of this sound wave.

Speed = _____ m/s

(2)

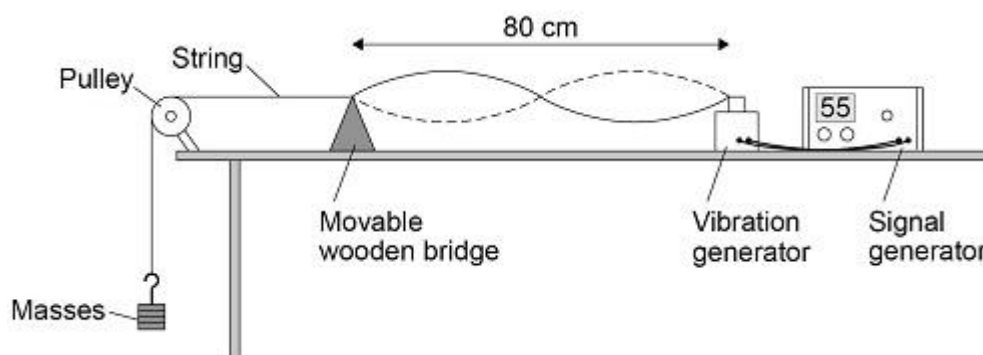
- (d) Sound waves are longitudinal. Describe a longitudinal sound wave.

(2)

(Total 6 marks)

Q3.

The following figure shows the apparatus used to investigate the waves in a stretched string.



The frequency of the signal generator is adjusted so that the wave shown in the figure is seen.

At this frequency the string vibrates between the two positions shown in the figure.

- (a) The wavelength of the wave shown in the figure above was measured as 80 cm
What piece of apparatus would have been suitable for measuring this wavelength?

_____ (1)

- (b) Write down the equation which links frequency, wavelength and wave speed.

_____ (1)

- (c) The string in the figure above vibrates at 55 Hz

Calculate the wave speed of the wave shown in the figure.

Use data given in the figure.

Wave speed = _____ m/s (3)

- (d) The frequency of the signal generator is increased.

This makes the wavelength of the wave change.

The wave speed stays the same.

Describe how the apparatus could be adjusted to show one complete wave without reducing the frequency.

(2)

- (e) A student wants to investigate how the speed of a wave on a stretched string depends on the tension in the string.

The student uses the apparatus in the figure above.

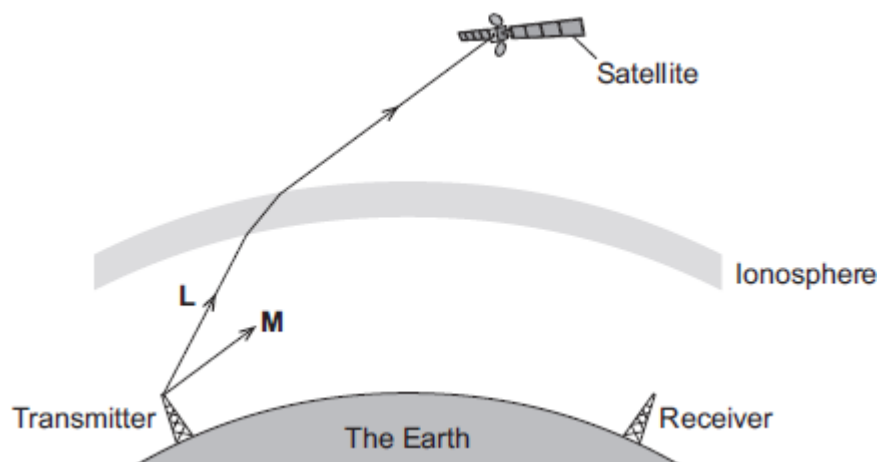
Describe a method the student could use for this investigation.

(4)
(Total 11 marks)

Q4.

Different parts of the electromagnetic spectrum are useful for different methods of communication.

The diagram shows a transmitter emitting two electromagnetic waves, **L** and **M**.



- (a) (i) Wave **L** is used to send a signal to a satellite.
Which part of the electromagnetic spectrum does wave **L** belong to?
- _____
- (1)
- (ii) What name is given to the process that occurs as wave **L** passes into the ionosphere?
- _____
- (1)
- (b) Wave **M** is **reflected** by the ionosphere.
- (i) On the diagram above, draw the path of wave **M** until it reaches the

receiver.

(2)

- (ii) On the diagram above, draw a line to show the normal where wave **M** meets the ionosphere. Label the line **N**.

(1)

- (c) Give **two** properties of all electromagnetic waves.

1. _____

2. _____

(2)

(Total 7 marks)

Q5.

X-rays and ultrasound can both be used for scanning internal organs.

- (a) Ultrasound is used to scan unborn babies but X-rays are **not** used to scan unborn babies.

Explain why.

(3)

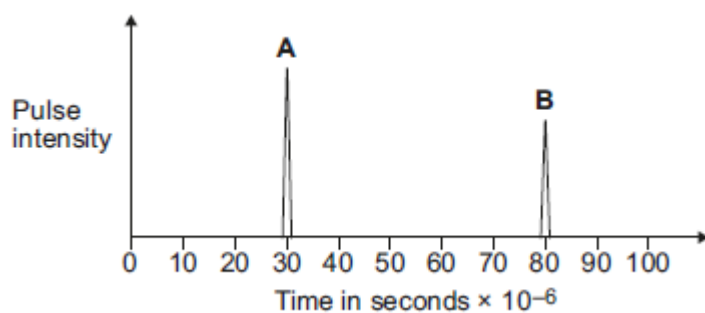
- (b) The behaviour of ultrasound waves when they meet a boundary between two different materials is used to produce an image.

Describe how.

(2)

- (c) **Figure 1** shows two pulses from a scan of an unborn baby. The emitted pulse is labelled **A**. The returning pulse picked up by the receiver is labelled **B**.

Figure 1



The closest distance between the unborn baby and the mother's skin is 4.0 cm. Use information from **Figure 1** to calculate the average speed of the pulse.

Average speed = _____ m/s

(3)

- (d) **Figure 2** shows an X-ray of an arm with a broken bone.

Figure 2



© emmy-images/iStock

- (i) Describe how X-rays are able to produce an image of bones.

(3)

(ii) Complete the following sentence.

X-rays are able to produce detailed images because their wavelength
is very _____ .

(1)

(Total 12 marks)

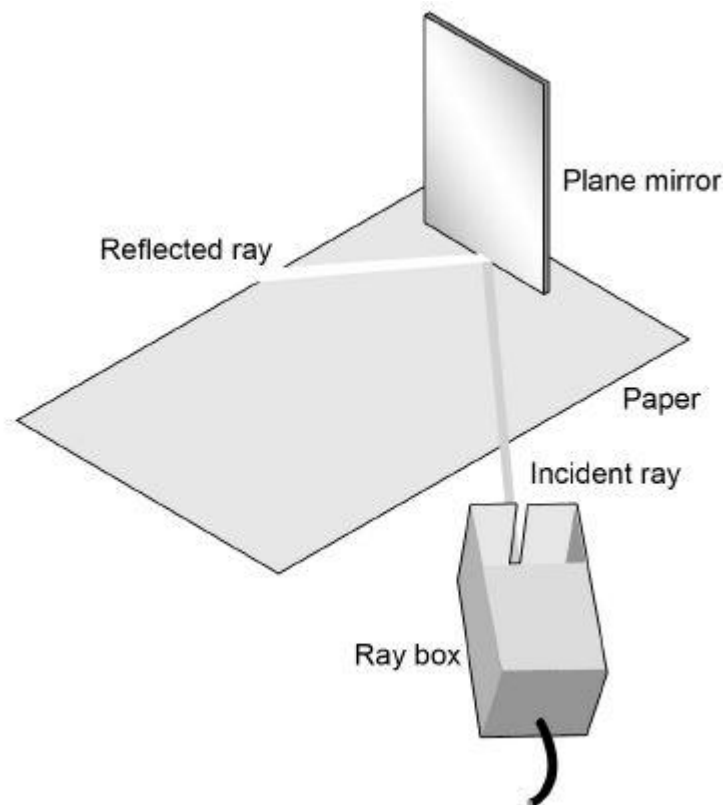
Q6.

The diagram below shows the apparatus a student used to investigate the reflection of light by a plane mirror.

The student drew four ray diagrams for each angle of incidence.

The student measured the angle of reflection from each diagram.

The table below gives the student's results.



Angle of incidence	Angle of reflection			
	Test 1	Test 2	Test 3	Test 4
20°	19°	22°	20°	19°
30°	31°	28°	32°	30°
40°	42°	40°	43°	41°
50°	56°	49°	53°	46°

- (a) For each angle of incidence, the angle of reflection has a range of values.

This is caused by an error.

What type of error will have caused each angle of reflection to have a range of values?

(1)

- (b) Suggest what the student may have done during the investigation to cause each angle of reflection to have a range of values.

(1)

- (c) Estimate the uncertainty in the angle of reflection when the angle of incidence is 50° .

Show how you determine your estimate.

Uncertainty = $\pm$ _____ $^\circ$
(2)

- (d) The student concluded that for a plane mirror, the angle of incidence is equal to the angle of reflection.

Explain whether you agree with this conclusion.

Use examples from the results in the table below in your answer.

(2)

- (e) What extra evidence could be collected to support the student's conclusion?

(1)

- (f) State **one** change the student should make to the apparatus if he wants to use the same method to investigate diffuse reflection.

(1)
(Total 8 marks)

Q7.

Different parts of the electromagnetic spectrum have different uses.

(a) The diagram shows the electromagnetic spectrum.

Radio waves	Microwaves	Infrared	Visible light	Ultraviolet	X-rays	Gamma rays
-------------	------------	----------	---------------	-------------	--------	------------



(i) Use the correct answers from the box to complete the sentence.

amplitude	frequency	speed	wavelength
------------------	------------------	--------------	-------------------

The arrow in the diagram is in the direction of increasing _____
and decreasing _____.

(2)

(ii) Draw a ring around the correct answer to complete the sentence.

The range of wavelengths for waves in the electromagnetic

spectrum is approximately _____ metres.

10^{-15}	to	10^4
10^{-4}	to	10^4
10^4	to	10^{15}

(1)

(b) The wavelength of a radio wave is 1500 m.
The speed of radio waves is 3.0×10^8 m / s.

Calculate the frequency of the radio wave.

Give the unit.

Frequency = _____

(3)

(c) (i) State **one** hazard of exposure to infrared radiation.

(1)

(ii) State **one** hazard of exposure to ultraviolet radiation.

(1)

(d) X-rays are used in hospitals for computed tomography (CT) scans.

(i) State **one** other medical use for X-rays.

(1)

(ii) State a property of X-rays that makes them suitable for your answer in part (d)(i).

(1)

(iii) The scientific unit of measurement used to measure the dose received from radiations, such as X-rays or background radiation, is the millisievert (mSv).

The table shows the X-ray dose resulting from CT scans of various parts of the body.

The table also shows the time it would take to get the same dose from background radiation.

Part of the body	X-ray dose in mSv	Time it would take to get the same dose from background radiation
Abdomen	9.0	3 years
Sinuses	0.5	2 months
Spine	4.0	16 months

A student suggests that the X-ray dose and the time it would take to get the same dose from background radiation are directly proportional.

Use calculations to test this suggestion and state your conclusion.

(3)
(Total 13 marks)

Mark schemes

Q1.

- (a) 20 000 Hz 1
- (b) 400 (m)
allow 1 mark for correct substitution ie 1600×0.25
provided no subsequent steps shown
an answer of 200 (m) gains 1 mark 2
- (c) twice 1
- (d) From pulse 1 to pulse 3 the distance (to the sea floor) decreased
accept the sea got shallower
or
the submarine went deeper for the distance decreased 1
- then (after pulse 3) the distance (to the sea floor) increased
accept the sea got deeper
or
the submarine rose for the distance increased
An answer of the distance decreased then increased gains 1 mark 1

[6]

Q2.

- (a) frequency 1
- (b) echo(es) 1
- (c) 340 (m/s)
allow 1 mark for correct substitution ie $25\,000 \times 0.0136$
provided no subsequent step
or
allow 1 mark for a correct calculation showing an incorrect value from conversion to hertz $\times 0.0136$
an answer of 0.34 gains 1 mark 2
- (d) (a wave where the) oscillations are parallel to the direction of energy transfer
both marking points may appear as labels on a diagram
accept vibrations for oscillations
accept in same direction as for parallel to

allow direction of wave (motion) for direction of energy transfer

allow 1 mark for a correct calculation showing an incorrect value from conversion to hertz $\times 0.0136$

1

causing (areas of) compression and rarefaction

accept correct description in terms of particles

mechanical wave is insufficient

needs a medium to travel through is insufficient

1

[6]

Q3.

(a) metre rule

allow metre ruler

allow tape measure

do not accept ruler

do not accept metre stick

1

(b) (wave) speed = frequency $\times$ wavelength

allow $v = f \lambda$

1

(c)

an answer of 44 (m/s) scores 3 marks

80 cm = 0.8 m

1

$v = 55 \times 0.8$

this mark may be awarded if wavelength is incorrectly or not converted

1

$v = 44$ (m/s)

allow correct calculation using an incorrectly or not converted wavelength

an answer of 4400 (m/s) scores 2 marks

1

(d) move the (wooden) bridge

1

to the right

dependent on 1st mp being scored

1

OR

change the mass/weight (on the string) scores 1 mark

add more masses/weights (to the string) scores both marks

- (e) **Level 2:** The design/plan would lead to the production of a valid outcome. All key steps are identified and logically sequenced.

3-4

Level 1: The design/plan would not lead to a valid outcome. Some relevant steps are identified, but links are not made clear.

1-2

No relevant content

0

Indicative content

add or take away masses from the string (ignore any stated values)

adjust frequency using the signal generator and/or move the wooden bridge

observe a steady / stationary pattern measure the wavelength

calculate wave speed from frequency and wavelength

a Level 1 answer should include a way of changing tension a complete Level 2 answer would include either changing frequency and/or moving the bridge

[11]

Q4.

- (a) (i) microwave

1

- (ii) refraction

1

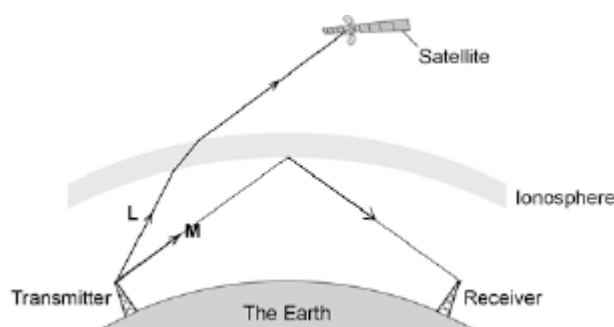
- (b) (i) wave M continues as a straight line to the ionosphere and shown reflected

accept reflection at or within the ionosphere

1

correctly reflected wave shown as a straight line reaching the top of the receiver

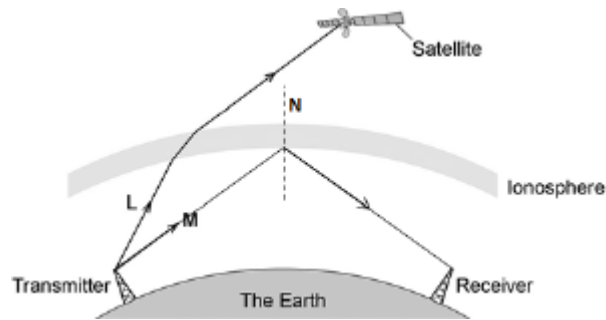
if more than 2 rays shown 1 mark maximum



ignore arrows

1

- (ii) normal drawn at point where their **M** meets the ionosphere



1

(c) any **two** from:

- transverse

- same speed (through air)

accept speed of light or $3 \times 10^8 \text{ m / s}$

- can be reflected

- can be refracted

- can be diffracted

- can be absorbed

- transfer energy

- can travel through a vacuum

an answer travel at the same speed though a vacuum scores 2 marks

- can be polarised

- show interference.

travel in straight lines is insufficient

2

[7]

Q5.

(a) ultrasound is not ionising

allow ultrasound does not harm the (unborn) baby

1

but X-rays are ionising

1

so X-rays increase the health risk to the (unborn) baby

accept specific examples of health risks, eg cancer, stunted growth, impaired brain function etc

X-rays are dangerous is insufficient

1

(b) ultrasound/waves are partially reflected

(when they meet a boundary) (between two different media / substances / tissues)

must be clear that not all of the wave is reflected

1

the time taken is measured (and is used to determine distances)

1

(c) 1600 (m/s)

800 (m/s) gains 2 marks

160 000 (m/s) gains 2 marks

0.0016 (m/s) gains 2 marks

allow 2 marks for

$$\frac{0.04}{25 \times 10^{-6}}$$

or

$$\frac{0.08}{50 \times 10^{-6}}$$

80 000 (m/s) gains 1 mark

0.0008 (m/s) gains 1 mark

allow 1 mark for

$$\frac{0.04}{25}$$

or

$$\frac{0.08}{50}$$

allow 1 mark for evidence of doubling the distance or halving the time

3

(d) (i) they are absorbed by bone

allow stopped for absorbed

X-rays are reflected negates this mark

1

they are transmitted by soft tissue

allow pass through for transmitted

allow flesh / muscle / fat

accept less (optically) dense material for soft tissue

1

(the transmitted) X-rays are detected

1

(ii) short

accept small

1

[12]

Q6.

(a) random

human error is insufficient

1

- (b) accept any practical suggestion that could cause a range of values
 e.g. misjudging the centre of the ray
 e.g. not replacing mirror / ray box in the same position
measuring the angle incorrectly is insufficient
moving the mirror / ray box is insufficient 1
- (c) range = 10
or
 mean of 51 calculated 1
- 5(°)
an answer of 5(°) scores 2 marks 1
- (d) within experimental accuracy the angle of incidence and the angle of reflection are the same
allow the angle of incidence is nearly the same as the angle of reflection
- or**
 the angle of reflection is usually different to the angle of incidence
allow only a few of the values are the same / similar
allow the idea of a range of values 1
- relevant use of data
 e.g.
 at 20° / 30° / 40° there is at least one measurement of angle of reflection that is exactly the same
or
 at 50° there are big differences
allow 50° includes anomalous results
an answer in terms of calculated mean(s) may score both marks
 e.g.
mean calculated for one or more angle of reflection (1)
conclusion correctly stating angle $i = / \neq$ angle r (1) 1
- (e) results could be collected for angles (of incidence) not yet measured
allow a stated angle of incidence e.g. 10° or 60°
changing the mirror is insufficient
ignore repeat the measurements 1
- (f) replace the mirror with an irregular reflecting surface
allow use an irregular reflecting surface
replace mirror with paper is insufficient
*do **not** accept use a glass block* 1

[8]

Q7.

- (a) (i) frequency 1
- wavelength 1
- (ii) 10^{-15} to 10^4 1
- (b) 2.0×10^5
- correct substitution of*
 3.0×10^8 / 1500 gains 1 mark 2
- Hz 1
- (c) (i) (skin) burns 1
- (ii) skin cancer / blindness 1
- (d) (i) any **one** from:
- (detecting) bone fractures
 - (detecting) dental problems
 - treating cancer
- 1
- (ii) any **one** from:
- affect photographic film
 - absorbed by bone
 - transmitted by soft tissue
 - kill (cancer) cells
- answer must link to answer given in (d)(i)* 1
- (iii) $9 / 36 = 0.25$
 $0.5 / 2 = 0.25$
 $4 / 16 = 0.25$
accept:
 $36 / 9 = 4$
 $2 / 0.5 = 4$
 $16 / 4 = 4$ 2
- conclusion based on calculation
- two calculations correct with a valid conclusion scores 2 marks*
- one correct calculation of k scores 1 mark* 1

[13]

Examiner reports

Q1.

- (a) Just under two thirds scored this mark for identifying the maximum frequency of human hearing.
- (b) The vast majority of students gained both marks for calculating the distance.
- (c) Just over two thirds answered this correctly.
- (d) A large number of students did not understand that the trend shown in the table needed to be identified. Only about a quarter of students gained both marks on this question. Under one fifth of students gained one mark, which was most commonly for realising that the distance decreased and increased again. A significant number of students talked about the time decreasing and increasing, without linking this to the distance between the submarine and the sea floor. A common response was to merely quote figures from the table, giving their answer in terms of time.

Q2.

Foundation

- (a) Two fifths of students scored this mark.
- (b) Less than a fifth of students scored a mark for this question. Many incorrect responses stated 'longitudinal', or another wave property like 'refraction' of 'diffraction.'
- (c) Almost four fifths of students scored 1 mark for the answer of '0.34' failing to convert 25.0 kHz into Hz. If a conversion was attempted it needed to be seen before the calculation otherwise it counted as a subsequent (incorrect) step, which means they scored zero.
- (d) Four fifths of students scored 0 marks for this question, most students believing that a longitudinal sound wave is a long sound wave. Most responses referred to amplitude, wavelength or frequency.

Higher

- (a) Three quarters of students scored a mark for this question.
- (b) Just under half the students scored a mark for this question. Many students, however, were baffled and gave an assortment of answers that ranged from wave behaviour (e.g. refraction) to wave properties (e.g. wavelength).
- (c) The vast majority of students scored 1 mark for the answer of '0.34'. Some students incorrectly rearranged the equation and scored zero. A few students realised that the frequency needed converting, but didn't do this correctly, but scored 1 mark for their final answer. One fifth of students scored 2 marks for the correct answer.
- (d) Lots of partial descriptions of 'waves parallel to energy transfer', without stating what was parallel, the word oscillation (or vibration) was needed for this mark. Compressions and rarefactions was more likely to be credited. A third of students scored 1 mark, but less than a tenth of students scored 2 marks.

Q3.

- (a) This question was successfully completed by 29% of the students. Both 'ruler' and 'metre stick' were common unacceptable answers. There were numerous other answers such as ammeter, oscilloscope and timer.
- (b) This equation was not well known with only 33% of the students scoring the mark.
- (c) Just over 60% of the students scored zero or did not attempt this question. A mark was awarded for converting the distance into metres, however this was rarely done.
- (d) Students could answer this in one of two ways, either with reference to the wooden bridge or with altering the mass hanging from string. Just over 23% of the students scored both marks, mainly for adding more masses as opposed to indicating which way the wooden bridge should be moved.
- (e) Nearly 20% of the students did not attempt this question. Many of those that did attempt the question gave a detailed written description of the apparatus shown in the diagram; this was not worth any marks. Those students that scored some marks identified the need to alter the mass on the string to adjust the tension. Some students correctly identified the need to measure the wavelength and then multiply this by the frequency as the way to determine the wave speed. However many students simply stated they would 'measure' the wave speed which was not creditworthy. Less than 1% of the students identified the need to move the bridge or adjust the frequency in order to see a wave pattern.

Q4.

- (a)
 - (i) One third of students scored the mark for this question. Many incorrect answers were seen, some being other waves in the electromagnetic spectrum, but many others were totally unconnected to electromagnetic waves.
 - (ii) The majority of students answered correctly identifying refraction as the process occurring as the wave passed through the ionosphere. Commonly seen incorrect answers were reflection and diffraction.
- (b)
 - (i) This question was well answered with about two thirds of students scoring 2 marks. Some students scored 1 mark only, usually for poor ray construction caused by drawing the ray freehand or the ray failing to reach the top of the receiver. Many students drew multiple rays which negated the 2nd mark. Some students drew the ray travelling to the satellite and back to the receiver, gaining no marks.
 - (ii) Over half of the students scored a mark. The mark was given for a dotted or a continuous line, labelled or not, as long as clearly the 'normal'. Common mistakes involved drawing a line perpendicular to the incident ray M. If the ray was refracted into the ionosphere, this mark still scored if the normal was correct. A significant number of students seemed unfamiliar with the term 'normal' and a range of different points or lines at various places within the diagram were labelled.
- (c) Around one quarter of students scored 2 marks, however many students gave more than 2 answers which included incorrect properties and negated their correct answers.

Q5.

- (a) The vast majority of students gained at least one mark, but less than half went on

to give a complete answer including a reference to the ionising properties of X-rays.

- (b) Just under half of students gained marks on this question. Many students understood that the sound reflected, but did not add that it was only partial reflection. The notion of time being recorded and used to calculate distance was only expressed by about one tenth of students.
- (c) This calculation caused problems for students with only a tiny minority managing to obtain the correct final answer. The majority of students neglected to either halve the time or double the distance from the mother's skin to the fetus.
- (d)
 - (i) A common misconception here was that bone reflected X-rays and the reflected X-rays were then detected. It was also not uncommon to see students stating that X-rays contained gamma rays or alpha particles. Many students who gained two marks neglected to mention that the X-rays passing through are detected. Around a quarter of students gained three marks.
 - (ii) This question was well answered with just over three quarters of students identifying why X-rays are able to produce detailed images.

Q6.

- (a) There were very few correct answers. Most answers described a specific error rather than a type of error. Many students incorrectly think that 'human error' is an acceptable answer.
- (b) There were many vague answers such as; 'knocked the ray box' that were not worthy of credit. Many students simply stated that there was a measuring error without further detail. Few students took the hint from the diagram that the rays were wide and so there would be difficulty in marking the exact position of each ray.
- (c) Nearly 70% of the students scored at least one mark for either calculating the mean or for calculating the range. Around 33% then calculated the uncertainty correctly.
- (d) Over 90% of the students scored at least one mark for either writing 'I agree / disagree' and quoting relevant data but not identifying the trend or for doing the converse.
- (e) About 23% of the students realised that the extra evidence would come from using a larger number of angles of incidence. Most answered in terms of repeating the same angles or comparing with another student's results.
- (f) Many students simply wrote 'change the mirror' but did not say for what. Others suggested replacing the mirror but with a non-reflecting surface or a glass block. In this question it was important to realise that the replacement was in the context of an investigation and not simply recalling the meaning of diffuse reflection.